Programming Assignment II: Summary

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Introduction

Implementation files

```
prefetcher/spp.l1d_pref             Task 1B: Stride Pattern Prefetcher
replacement/irrip.l2c_repl          Task 2A: I-RRIP
replacement/dyn_irrip.l2c_repl      Tasks 2B and 3: Dynamic I-RRIP
```

Methodology

- Every provided trace is simulated with 25M warmup and 25M simulation instructions.
- The baseline has no L1D prefetcher and LRU replacement at L2C. Speedup is IPC divided by the baseline's IPC on the same trace.

Task 1

Stride Pattern Prefetching

Task 1A: IP-stride prefetcher

Task 1B: Design

Storage overhead

Accuracy throttling

Results

Task 2A

Instruction-Conscious L2 Replacement Policy (I-RRIP)

Design: insertion, promotion, eviction

Results

Task 2B

Dynamic I-RRIP

Leader sets and decision logic

Sensitivity study

Results

Task 3

Incorporating SPP with Dynamic I-RRIP

Handling prefetched lines

Results